

Madhav Khanal

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SUMMARY

AI/ML Developer and Data Scientist with experience in machine learning projects. Skilled in Python, C, Java, FastAPI, and TensorFlow/Keras. Passionate about building AI-powered applications and delivering production-ready solutions.

WORK EXPERIENCE

Data Science Intern (Training Program) — Fusemachines Nepal

2024

Performed data cleaning, preprocessing, and exploratory data analysis (EDA) on real-world datasets using Python. Built and evaluated machine learning models using statistical and supervised learning techniques, applied feature engineering, and visualized insights to support data-driven decision making. Collaborated in an industry-oriented environment to understand business problems and translate them into data science solutions.

PROJECTS

Trinetra – Automated Security Vulnerability Detection

[Link to Demo](#)

An AI-powered platform developed to identify and remediate security vulnerabilities in C programs. The system features a full-stack architecture using React and an asynchronous FastAPI backend with secure authentication and admin management. The platform also supports automated PDF report generation, scan history tracking, and subscription-based usage.

Plant Disease Detection Platform

Developed an end-to-end web-based application for automated plant disease detection using leaf images. Users upload leaf photos via a React-based frontend, which communicates with a Django backend for image processing and prediction. A CNN-based machine learning model analyzes the images to accurately identify plant diseases. The system integrates frontend, backend, and ML components through RESTful APIs to deliver real-time disease detection.

EDUCATION

2021 - 2025	Bachelor of Software Engineering Degree at Nepal College of Information Technology (GPA: 3.57/4.0)	
2020	Class 12th NEB	(3.47)
2017	Class 10th NEB	(3.60)

SKILLS

Programming Languages	C, C++, Python, Java
Web Technologies	FastAPI, REST APIs, TailwindCSS, HTML, CSS
AI / Machine Learning	CNNs, Transformers, Deep Learning, Model Evaluation, Data Preprocessing
Frameworks & Libraries	TensorFlow, Keras, PyTorch, SQLAlchemy, Pydantic